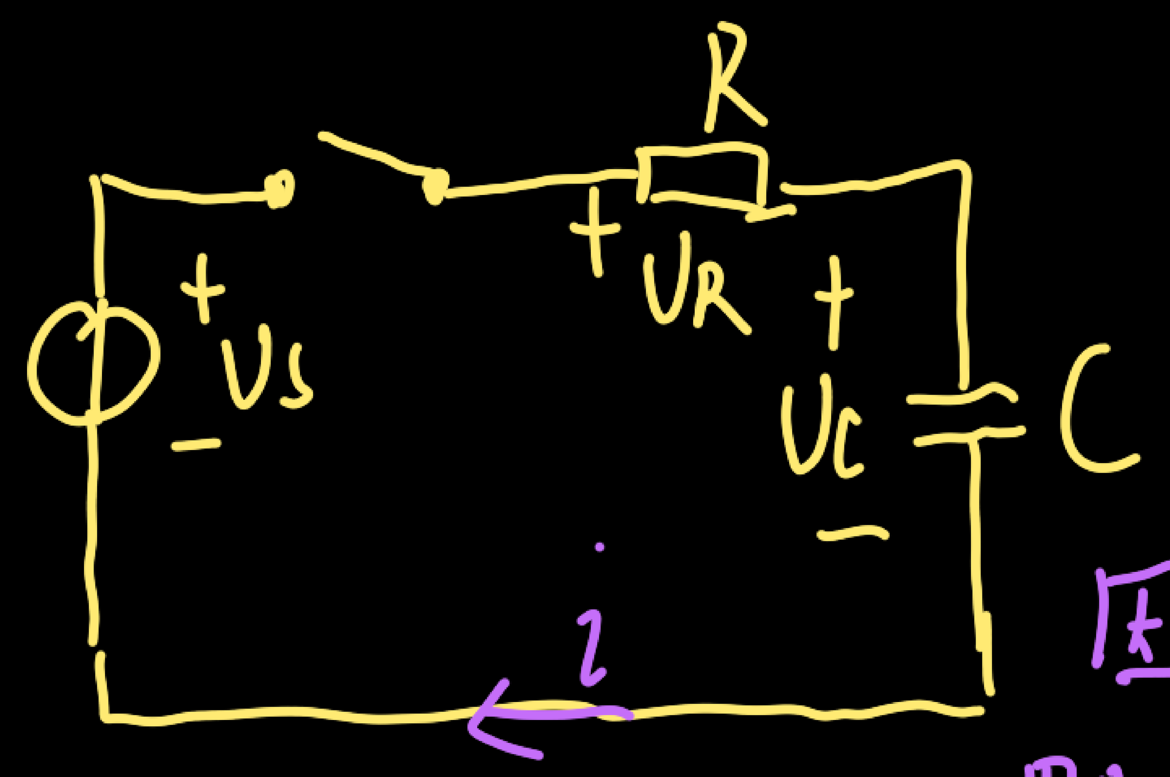

$$RC\frac{du_C}{dt} + u_C = U_S \qquad u_C = u_C' + u_C''$$

因为 U_S 为常数，故非齐次特解 $u_C'' = C_0$

则 $0 + C_0 = U_S \qquad u_C'' = U_S$

$$u_C' = Ae^{-\frac{t}{RC}} \qquad u_C(0_-) = u_C(0_+) = Ae^{-\frac{0}{RC}} + U_S = 0 \qquad A = -U_S$$

则 $u_C = -U_S e^{-\frac{t}{RC}} + U_S = U_S\left(1 - e^{-\frac{t}{RC}}\right)$

$$i = C\frac{du_C}{dt} = -U_S C \cdot -\frac{1}{RC} e^{-\frac{t}{RC}} = \frac{U_S}{R} e^{-\frac{t}{RC}}$$

电源对电路的控制

非齐次特解 $\longrightarrow U_S \longrightarrow$ 强制分量、稳态分量

齐次通解 $\longrightarrow RC\frac{du_C}{dt} + u_C \longrightarrow$ 自由分量、暂态分量

↓

电路除电源内部的互相影响